

Validity and Reliability of the Korean Version of the Motivation to Change Lifestyle and Health Behaviors for Dementia Risk Reduction Scale

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Abstract

Objective: This study aimed to translate and validate the Motivation to Change Lifestyle and Health Behaviors for Dementia Risk Reduction (MCLHB-DRR) scale among middle-aged and older Korean adults.

Methods: This study included a sample of Korean adults aged ≥ 50 years recruited from an online panel. Content and construct validity were examined to determine the scale's validity. Internal consistency and test-retest reliability were examined to determine its reliability.

Results: A total of 270 participants completed the questionnaire. Content validity was determined by professors and researchers of occupational therapy. One item was deleted during the content validity assessment. The confirmatory factor analysis results showed a good model fit ($\chi^2/\text{degree of freedom} = 1.845$, comparative fit index = 0.939, Tucker-Lewis index = 0.929, root mean square error of approximation = 0.056). Cronbach's alpha values were 0.821 - 0.900, indicating good internal consistency. The test and retest scores did not differ significantly.

Conclusion: The Korean version of the MCLHB-DRR scale has validity and reliability for use among the Korean population, to assess beliefs and attitudes toward lifestyle change to reduce the risk of dementia.

Key words: Assessment, Dementia risk reduction, Health behaviors, Lifestyle, Motivation

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I . Introduction

Dementia is a major public health concern affecting approximately 55 million individuals worldwide, with numbers expected to reach 152 million by 2050 (WHO, 2023). In South Korea, a 60% (12.98 million) increase in the older adult population is projected by 2030, and older adults with dementia are expected to increase by 64% (1.36 million), indicating a faster dementia growth than the overall older adult population (Ministry of Health and Welfare, 2020). This rapid increase in patients with dementia results in higher healthcare costs and substantial socioeconomic burdens on families and caregivers (Lee & Seong, 2018), necessitating proactive national responses (Wimo et al., 2017).

Globally, efforts are being made to implement policies aimed at reducing the burden of dementia. The World Health Organization (WHO, 2012) and the G8 Dementia Summit (Global Action Against Dementia, 2013) emphasized prevention as a critical strategy. The WHO (2019) guidelines on risk reduction highlight managing modifiable risk factors to delay or slow down the onset or progression of dementia. Studies have shown a relationship among cognitive impairment, dementia, and lifestyle-related risk factors, such as physical inactivity, smoking, hypertension, obesity, diabetes, and depression (Deckers et al., 2015; Livingston et al., 2017; Norton et al., 2014). The Lancet Commission reported that 40% of dementia cases can be attributed to 12 modifiable risk factors, including physical inactivity, diabetes, smoking, depression, low social contact, and excessive alcohol consumption (Livingston et al., 2020).

While identifying modifiable risk factors for dementia is crucial, practical interventions are

necessary. Large randomized controlled trials, such as the Finnish Geriatric Intervention Study to Prevent Cognitive Impairment and Disability (FINGER), Multidomain Alzheimer Preventive Trial (MAPT), and Prevention of Dementia by Intensive Vascular Care (PreDIVA), have focused on multidomain lifestyle interventions for older adults at risk (Ngandu et al., 2015; van Charante et al., 2016; Vellas et al., 2014). These trials demonstrated the potential to reduce the risk of developing dementia (Kivipelto et al., 2013).

In addition to international studies, Korean studies, such as the “South Korean study to Prevent cognitive impairment and protect BRAIN health through lifestyle intervention in at-risk elderly people (SUPERBRAIN)” (Lee et al., 2020; Moon et al., 2021; Park et al., 2020) based on the FINGER model, demonstrated the feasibility of adapting such interventions to Korea’s cultural and situational contexts (Moon et al., 2022). These studies highlight modifiable risk factors and the effectiveness of preventive measures within the Korean population.

Furthermore, South Korea has developed the “Guidelines for Dementia Prevention 3.3.3”, advocating exercise, balanced meals, and reading, while discouraging smoking and excessive alcohol, and recommending regular health checkups and early dementia screening (Ministry of Health and Welfare & Central Dementia Center, 2014). Despite active promotion, adherence among older adults is low, scoring 40–63 out of 100 (Cho, 2019; Kim & Yang, 2016; Ryu & Lee, 2022). Attitudes toward dementia among individuals in their 50s and 60s show lower compliance compared with other age groups, highlighting the need for targeted dementia awareness and educational programs for this age group (Central Dementia Center & Myongji Hospital, 2021). Therefore, factors influencing the lifestyle

and health behavioral changes should be identified to reduce the risk of dementia.

Social cognitive theories highlight that health behavioral changes involve a complex cognitive process influenced by various factors, including beliefs and attitudes about behavioral changes, the individual's current stage of change, reinforcement management, perceived benefits, and barriers (Glanz, 2016; Joxhorst et al., 2020). Measuring beliefs and attitudes toward lifestyle changes that reduce the risk of dementia is crucial, as it helps predict individuals' willingness to adapt their lifestyles and behaviors (Joxhorst et al., 2020).

The Health Belief Model (HBM) is widely used to explain health behaviors and factors that motivate or deter individuals from engaging in behavioral changes (Glanz et al., 2002; Kim et al., 2014; Sayegh & Knight, 2013). This model further indicates that health-promoting behaviors more likely change when a person perceives a health threat through perceived susceptibility and severity and believes that the benefits of changes outweigh the barriers (Kim et al., 2014). Cues to action, health motivation, and self-efficacy are also essential (Janz et al., 2002). The HBM has been used both internationally and domestically to predict health-related behaviors, including dementia prevention (Jo et al., 2004; Kim & Chang, 2020; Lee et al., 2014). Studies have reported various beliefs and motivational factors promoting or inhibiting dementia prevention behaviors, such as screening, intention to engage in preventive behaviors, and actual preventive behaviors (Choi et al., 2019; Oh, 2017; Seo & Choi, 2021; Yoo & Kim, 2017).

Based on the HBM, Kim et al. (2014) developed the Motivation to Change Lifestyle and Health Behaviors for Dementia Risk Reduction (MCLHB-DRR) scale was developed in Australia. This scale includes 27 items across seven HBM subscales

and measures beliefs and attitudes toward health behaviors and lifestyle changes aimed at reducing dementia risks. It has been proven reliable and valid among diverse populations in Australia, Turkey, the Netherlands, Israel, and China (Joxhorst et al., 2020; Kim et al., 2014; Lin et al., 2023; Shvedko et al., 2023; Zehirlioglu et al., 2019). However, its applicability and effectiveness in the culturally distinct context of Korea have not yet been explored. A meta-analysis of HBM studies in Korea reveals that the key variables of the model do not consistently correlate with health behaviors and vary depending on the type of health behavior, disease, and respondent characteristics. The influence of HBM variables can vary by race, region, or environment, indicating that health beliefs and behaviors may change with environmental factors (Jung, 2013; Lee et al., 2014).

II. Methods

1. Participants and Data Collection

This was a psychometric study evaluating the validity and reliability of the MCLHB-DRR scale in South Korea. Based on previous scale validity and reliability studies, the sample size suggested was five times the number of items in the scale (Sousa & Rojjanasrirat, 2011; Zehirlioglu et al., 2019). An estimated sample size of 135~270 represents 5~10 times the 27 items in the scale.

A total of 270 individuals were recruited between September and October 2023 using Embrain (www.embrain.com). Embrain holds a sizable online panel with approximately 1.2 million members. The inclusion criteria included community-dwelling

adults aged ≥ 50 years. The exclusion criteria included individuals diagnosed with dementia or other psychiatric disorders.

Data were collected using a sociodemographic questionnaire created by the researchers and the MCLHB-DRR scale. The sociodemographic questionnaire collected data on age, sex, marital status, education level, employment status, area of residency, having relatives or friends with dementia, and providing care for relatives or friends with dementia and was administered before completion of the MCLHB-DRR scale.

The test-retest reliability of the MCLHB-DRR scale was assessed by re-administering it to a subsample of 50 participants from the original 270 participants, within 3 weeks of the first application (Streiner & Kottner, 2014). All participants had previously given consent to be recontacted for this purpose.

2. The MCLHB-DRR Scale

The original MCLHB-DRR scale includes 27 items covering seven HBM subscales on lifestyle and health behavioral changes to reduce the dementia risk: perceived susceptibility (four items), perceived severity (five items), perceived benefits (four items), perceived barriers (four items), cues to action (four items), general health motivation (four items), and self-efficacy (two items). All items are scored on a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree), with higher scores indicating higher motivation to change lifestyle and health behaviors to reduce dementia risk. The MCLHB-DRR questionnaire has been reported to have good internal consistency (Cronbach's alpha of 0.608 - 0.864), and all subscales have moderate test-retest reliability (Cronbach's alpha of 0.552~

0.776) (Kim et al., 2014).

3. Ethical Considerations

This study was approved by the Institutional Review Board of the author's institution (IRB No. 1041849-202308-SB-149-02), and written informed consent was obtained from all participants before participation. Participants were required to read and consent to an information sheet before completing the questionnaire. No personal information was obtained from the participants. The data were coded anonymously and protected using passwords.

4. Scale Translation

Permission was obtained from the original author of the MCLHB-DRR scale for translation and use in the Korean context before starting the primary translation. The WHO guidelines for the translation and adaptation of instruments were followed to ensure the validity of the translation procedure (WHO, 2016). The translation process was conducted in the following order: forward translation, expert panel review, back-translation, pretesting and cognitive interviewing, and completion of the final translated version (Figure 1).

First, a researcher and professor from the Department of Occupational Therapy conducted the forward translation. Second, an expert panel reviewed this translated version for accuracy and understandability, including three health experts proficient in both English and Korean who had experience in instrument translation or development. Third, a native English-speaking health expert who was blinded of the instrument back-translated the preliminary translated Korean version of the MCLHB-DRR into English. Differences in expressions

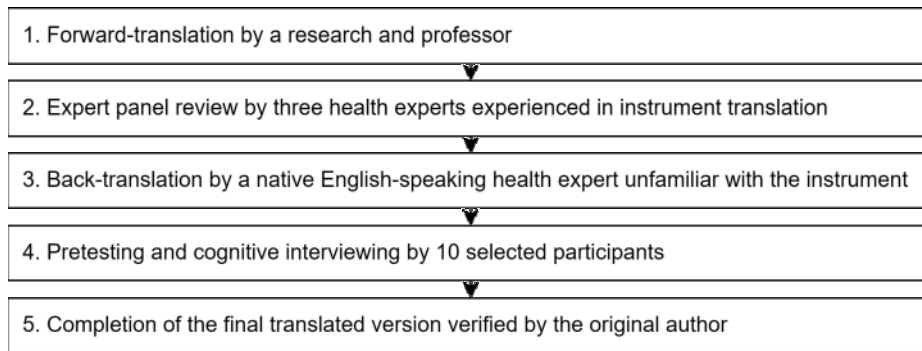


Figure 1. Scale Translation Process

and vocabulary choices were reviewed and corrected through discussions between the primary and back-translators. Finally, the revised back-translated version was verified by the original developer to confirm the absence of disparities between the back-translated and original versions.

After creating the prefinal version of the questionnaire, preliminary testing for clarity of expression was performed with ten participants who met the participant selection criteria for this study. These participants were excluded from the statistical analysis. The number of items and scale forms remained consistent with those of the original instrument. The mean completion time was approximately 10 min. The pretest results showed that the participants had no problems understanding and responding to the questionnaire. Finally, the original author was asked to review the translation, and the final translated version of the scale was completed.

5. Content Validity

A group of 13 experts in the dementia healthcare field was selected and informed about the study purpose and content validity assessment process. Of the 13 experts, seven were faculty members from the Department of Occupational Therapy

and Health, four were postdoctoral researchers majoring in occupational therapy, and two were doctoral students with clinical experience in occupational therapy. The criteria for selecting experts were (1) individuals with at least 1 year of research experience and (2) individuals with experience in dementia-related research. General information about the experts who participated in the content validity verification process is listed in Table 1.

Based on the technique presented by Lynn (1986), the experts were asked to rate the relevance and clarity of each item on a 4-point Likert scale ranging from 1 (not relevant) to 4 (very relevant and succinct) to calculate the content validity index (CVI). The number of experts who scored 3 or 4 was divided by the total number of experts to calculate the CVI. According to Lynn (1986), an item's CVI (item-level CVI; I-CVI) should be 1.00 among 3~5 experts and ≥ 0.78 among 6~10 experts to consider the I-CVI acceptable.

6. Statistical Analysis

Baseline descriptive statistics were calculated for sociodemographic characteristics and the MCLHB-DRR questionnaire. The construct validity of the MCLHB-DRR questionnaire was evaluated

Table 1. General Characteristic of the Experts Involved in the Content Validity Study (N = 13)

Characteristic		n (%)
Gender	Male	6 (46.2)
	Female	7 (53.8)
Age	20~29 years	1 (7.7)
	30~39 years	8 (61.5)
	40 years or older	4 (30.8)
Occupation	Graduate Student	2 (15.4)
	Postdoctoral Researcher	4 (30.8)
	Professor	7 (53.8)
Clinical or Research Experience	Less than 5 years	1 (7.7)
	5~9 years	6 (46.2)
	10 years or more	6 (46.2)

The sum of the percentages does not equal 100% because of rounding

using confirmatory factor analysis (CFA). CFA is a data analysis method that validates general construct concepts (latent variables) from observable measurement items (observed variables) (De Vet et al., 2011). Confirmatory factor analysis is conducted when the construct concepts and their measurement items are predetermined before the analysis (Polit & Yang, 2016). CFA with the maximum likelihood method was used to examine model fit. Multiple fit indices were used to evaluate the goodness-of-fit model: chi-square divided by degree of freedom (χ^2/df), root mean square error of approximation (RMSEA), Tucker-Lewis index (TLI), and comparative fit index (CFI). An acceptable model fit was defined as $\chi^2/df < 3$ (Hu & Bentler, 1999), RMSEA < 0.05 (excellent) to < 0.09 (moderate; Hair, 2014), TLI > 0.90 (moderate) to > 0.95 (excellent), and CFI > 0.90 (moderate) to > 0.95 (excellent; Hu & Bentler, 1999). Factor loadings were examined. Each item was considered satisfactory if its loading was > 0.45 (Byrne, 2010).

The internal consistency of the subscales was evaluated using correlation analysis (Cronbach's alpha and item-total correlation). Cronbach's alpha of 0.7 indicated satisfactory internal consistency

(Field, 2018). An item was considered for deletion if its item-total correlation fell < 0.30 (De Vet et al., 2011). Test-retest reliability was analyzed using Pearson's correlation and paired-samples t-test. The correlation coefficient between the test and retest was at least 0.20 (Nunnally & Bernstein, 2010; Rattray & Jones, 2007). Before conducting the Pearson correlation analysis and the paired-samples t-test, the collected data were confirmed to follow a normal distribution.

Descriptive statistics were analyzed using SPSS software (version 27.0; SPSS Inc., Chicago, IL, USA). CFA was performed using AMOS version 25.0 to determine the scale's factor structure. A p value < 0.05 indicated statistical significance.

III. Results

1. Participants and Data Collection

Table 2 presents the participant's sociodemographic characteristics. A total of 270 participants were included in this study. The participant's mean age (standard deviation) was 58.4 (5.5) years, 50%

Table 2. General Characteristics of the Study Participants

(N = 270)

Characteristic	Participants (n = 270)
Age, years (<i>mean ± SD</i>)	58.4 ± 5.5
Gender, n (%)	
Male	135 (50.0)
Female	135 (50.0)
Marital status, n (%)	
Married	226 (83.7)
Single	17 (6.3)
Divorced	19 (7.0)
Widow/widower	8 (3.0)
Education level, n (%)	
Less than or equal to high school	80 (29.6)
University	161 (59.6)
Graduate school or above	29 (10.7)
Currently working, n (%)	175 (64.8)
Residential area size, n (%)	
Metropolitan area	145 (53.7)
Provincial city	56 (20.7)
Midsized city	64 (23.7)
Rural area	5 (1.9)
Subjective health status, n (%)	
High	106 (39.3)
Middle	124 (45.9)
Low	40 (14.8)
Chronic disease, n (%) ^a	
None	124 (45.9)
Hypertension	73 (27.0)
Diabetes	36 (13.3)
Hyperlipidemia	74 (27.4)
Arthritis	18 (6.7)
Others	14 (5.2)
Have relatives/friends with dementia, n (%)	86 (31.9)
Providing care for relatives/friends with dementia, n (%)	41 (15.2)

The sum of the percentages does not equal 100% because of rounding

SD: Standard Deviation

^aMultiple responses were allowed for chronic diseases

were women, and 83.7% were married. Additionally, more than half of the participants completed a university education (59.6%), 29.6% had a high school or lower education level, and 10.7% had a

graduate or higher education level. Furthermore, 64.8% of the participants were employed, and 53.7% lived in metropolitan areas. Of the participants, 31.9% reported having relatives or

friends with dementia, and 15.2% reported providing care for relatives or friends with dementia. Additionally, the subjective health status and presence of chronic diseases were assessed. The most common response for subjective health status was “middle”, comprising 45.9% of the responses. Regarding objective health status that reflects the presence of chronic diseases, 55.1% of participants reported having at least one chronic disease.

2. Validity Analyses

1) Content Validity

The I-CVIs of the 27 items ranged from 0.6 to 1. In this study, with 13 experts, an I-CVI $\geq$ 0.78 was required for an item to be considered satisfactory regarding content validity (Lynn, 1986). Item 8 had an I-CVI of 0.6 and was removed. The I-CVI values for each item are presented in Table 3.

2) Construct Validity

The fit of the model was assessed using CFA with the maximum likelihood method (Table 4). The initial analysis of the seven-factor model with all 27 items (Model 1) indicated a moderate fit (χ^2/df = 2.038; CFI = 0.910; TLI = 0.923; RMSEA = 0.062). The analysis of the seven-factor model without Item 8 (Model 2) (Figure 2) revealed a good fit (χ^2/df = 1.845; CFI = 0.929; TLI = 0.939; RMSEA = 0.056), indicating that Model 2 has a better fit

to the data than Model 1. The factor loadings for Model 2 ranged from 0.588 to 0.911 (Table 5).

Table 3. Item-Level Content Validity Index (I-CVI) Scores for Each Item (N = 13)

Item	Mean $\pm$ SD	I-CVI
Q1	3.62 $\pm$ 0.65	0.92
Q2	3.46 $\pm$ 0.78	0.85
Q3	3.31 $\pm$ 0.75	0.85
Q4	3.31 $\pm$ 0.75	0.85
Q5	3.77 $\pm$ 0.44	1.00
Q6	3.46 $\pm$ 0.78	0.85
Q7	3.54 $\pm$ 0.52	1.00
Q8	3.08 $\pm$ 0.95	0.62
Q9	3.54 $\pm$ 0.66	0.92
Q10	3.54 $\pm$ 0.66	1.00
Q11	3.85 $\pm$ 0.38	0.92
Q12	3.62 $\pm$ 0.65	0.92
Q13	3.85 $\pm$ 0.55	1.00
Q14	3.62 $\pm$ 0.51	1.00
Q15	3.85 $\pm$ 0.38	1.00
Q16	3.69 $\pm$ 0.48	1.00
Q17	3.85 $\pm$ 0.38	1.00
Q18	3.69 $\pm$ 0.48	1.00
Q19	3.85 $\pm$ 0.38	1.00
Q20	3.77 $\pm$ 0.44	1.00
Q21	3.69 $\pm$ 0.63	0.92
Q22	3.77 $\pm$ 0.44	1.00
Q23	3.62 $\pm$ 0.51	1.00
Q24	3.77 $\pm$ 0.44	1.00
Q25	3.77 $\pm$ 0.44	1.00
Q26	3.85 $\pm$ 0.38	1.00
Q27	3.62 $\pm$ 0.65	0.92

SD: Standard Deviation

Table 4. Goodness-of-Fit Indices for the Korean MCLHB-DRR Model

	χ^2/df	TLI	CFI	RMSEA
Korean MCLHB-DRR Model 1	2.038	0.910	0.923	0.062
Korean MCLHB-DRR Model 2	1.845	0.929	0.939	0.056

Model 1 is a seven-factor model with all 27 items, and Model 2 is a seven-factor model with 26 items (without Item 8)

RMSEA: Root Mean Square Error of Approximation, CFI: Comparative Fit Index, TLI: Tucker-Lewis Index

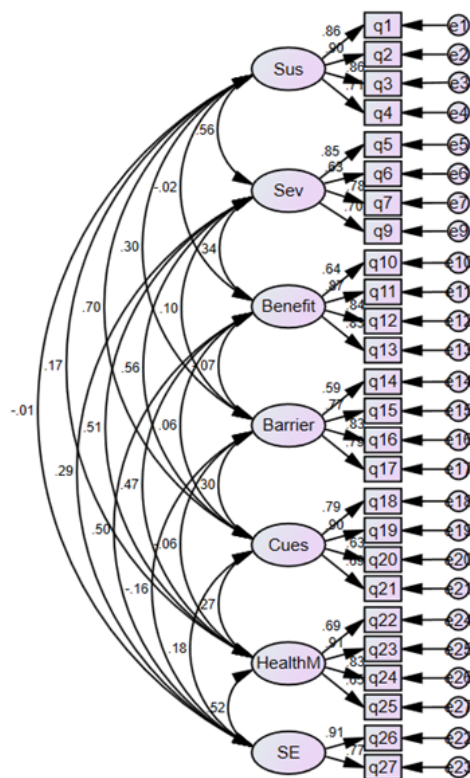


Figure 2. Confirmatory Factor Analysis Model with the 26-item (without item 8)

Sus: Perceived Susceptibility, Sev: Perceived Severity, Benefit: Perceived Benefits, Barrier: Perceived Barriers, Cues: Cues to Action, HealthM: General Health Motivation, SE: Self-Efficacy

3. Reliability Analyses

1) Internal Consistency

The item-total correlation analysis revealed a positive correlation between the items and the total MCLHB-DRR questionnaire. However, Items 12 ($r = 0.293$), 13 ($r = 0.298$), 14 ($r = 0.278$), 15 ($r = 0.204$), 16 ($r = 0.295$), 17 ($r = 0.257$), and 27 ($r = 0.240$) had correlations of < 0.30 , which were considered nonhomogeneous with other items in the scale and could be removed. However, all items except Item 8 were retained because of their high content validity (I-CVI of 0.9). This decision was supported by the strong thematic

Table 5. Confirmatory Factor Analysis Report

Subscales	Item	Factor Loading ^a
Perceived susceptibility	Q1	0.859*
	Q2	0.898*
	Q3	0.861*
	Q4	0.714*
Perceived severity	Q5	0.854*
	Q6	0.634*
	Q7	0.777*
	Q9	0.696*
Perceived benefits	Q10	0.636*
	Q11	0.874*
	Q12	0.841*
	Q13	0.832*
Perceived barriers	Q14	0.588*
	Q15	0.770*
	Q16	0.833*
	Q17	0.791*
Cues to action	Q18	0.791*
	Q19	0.900*
	Q20	0.631*
	Q21	0.687*
General health motivation	Q22	0.911*
	Q23	0.774*
	Q24	0.694*
	Q25	0.906*
Self-efficacy	Q26	0.832*
	Q27	0.649*

^aAll factor loadings are significant at $p < 0.001$

relevance of these items to the overall construct being measured, underscoring their importance despite lower correlations.

Table 6 shows the MCLHB-DRR subscales and their Cronbach's alpha values. Cronbach's alpha values were 0.82~0.900 for self-efficacy and perceived susceptibility, indicating good internal consistency. Cronbach's alpha values were 0.900, 0.821, 0.870, 0.833, 0.836, 0.841, and 0.827 for perceived susceptibility, severity, benefits, barriers, cues to action, general health motivation, and

Table 6. Internal Consistency Analysis for the Subscales (N = 270)

Subscales	No. of items	Range of scores	Mean (SD)	Cronbach's alpha
Perceived susceptibility	4	4-20	9.3 (3.0)	0.900
Perceived severity	4 ^a	4-20	13.5 (3.4)	0.821
Perceived benefits	4	4-20	15.8 (2.3)	0.870
Perceived barriers	4	4-20	11.1 (2.9)	0.833
Cues to action	4	4-20	11.1 (3.1)	0.836
General health motivation	4	4-20	15.9 (2.4)	0.841
Self-efficacy	2	1-10	7.3 (1.4)	0.827

SD: Standard Deviation

^aItem 8 was deleted

Table 7. Test-Retest Reliability Analysis for the Subscales (N = 50)

Subscales	Test ^a	Retest ^a	<i>t</i>	<i>r</i>	<i>p</i>
Perceived susceptibility	9.22 (2.53)	9.08 (2.28)	0.522	0.694	0.000
Perceived severity	13.94 (2.88)	13.64 (2.95)	0.690	0.446	0.001
Perceived benefits	16.12 (1.72)	15.82 (1.73)	1.364	0.595	0.000
Perceived barriers	11.36 (2.55)	11.06 (2.68)	1.019	0.684	0.000
Cues to action	11.32 (2.57)	10.88 (2.96)	1.265	0.613	0.000
General health motivation	15.86 (1.98)	15.96 (1.63)	-0.359	0.416	0.003
Self-efficacy	7.30 (1.09)	7.12 (1.47)	0.860	0.359	0.010

^aMean (standard deviation)

self-efficacy, respectively.

2) Test-Retest Reliability

A statistically significant positive correlation was observed between the test and retest scores. The test-retest reliability coefficients of the subscales were positive ($p > 0.20$; $p < 0.05$). Thus, no significant difference was observed between the test and retest mean scores of the MCLHB-DRR scale (Table 7).

IV. Discussion

This study presented the Korean version of the MCLHB-DRR scale, consisting of 26 items, and validated its use for measuring beliefs and

attitudes toward lifestyle and health behavioral changes to reduce dementia risk in adults aged ≥ 50 years.

First, an expert panel reviewed the content validity of the Korean version of the MCLHB-DRR scale, a common method for determining measurement tool validity (Zehirlioglu et al., 2019). The I-CVIs for the 27 items ranged from 0.85 to 1, except for item 8, which garnered an I-CVI of 0.6. Item 8 was removed as it did not meet the 0.78 threshold for satisfactory content validity, likely due to cultural differences in the perception of “fear of dementia” between Australian and Korean residents.

Item 8 pertains to “Perceived severity,” a subconstruct of the HBM, referring to how anxious and stressed individuals would feel if they developed dementia. Lee and Jung (2018) identified four

dimensions of fear of dementia among older Koreans—cognitive, emotional, social, and behavioral—with 14 attributes and 34 indicators. In the behavioral dimension, indicators such as increased sweating, heart rate, breathing rate, and sleep disturbances are highlighted, but not nausea. During the content validity process for the Korean version of the MCLHB-DRR scale, the majority of 13 experts agreed that “feeling nauseous” did not match Korean cultural perceptions of dementia’s severity. The other items under perceived severity, excluding item 8, aligned well with the attributes of dementia fear among older Koreans. These include emotional-negative feelings, behavioral-peripheral autonomic nervous system responses, cognitive impossibility for control, and emotional-pessimistic thoughts. Therefore, item 8 was removed, and the remaining 26 items were retained.

For construct validity, the CFA showed that the 26-item model (without item 8) fit the data better than the 27-item model. The value of $\chi^2/df = 1.845$ was lower than those for the Dutch ($\chi^2/df = 2.130$), Israeli ($\chi^2/df = 2.146$), Australian ($\chi^2/df = 2.379$), and Chinese populations ($\chi^2/df = 2.14$), but similar to the Turkish population ($\chi^2/df = 1.818$). The CFI and RMSEA values of 0.929 and 0.056 were higher than those of the original Australian scale, respectively (CFI = 0.920, RMSEA = 0.047). A χ^2/df ratio ≤ 2 is considered a superior fit (Alavi et al., 2020; Cole, 1987). CFA can determine an instrument’s validity for use in different cultures (Harrington, 2009). In this study, CFA results validated the factor structure of the scale, confirming its applicability to the Korean population.

The reliability analysis revealed that Cronbach’s alpha values from 0.821 to 0.900 indicate good internal consistency, higher than the original

scale (0.608 to 0.864) (Kim et al., 2014). Test-retest reliability showed significant positive correlations between scores > 3 weeks ($p < 0.05$), with coefficients > 0.20 (Nunnally & Bernstein, 2010; Rattray & Jones, 2007). These results are similar to those of the Turkish version of the MCLHB-DRR (Zehirlioglu et al., 2019), confirming that respondents had consistent reactions to the items, indicating acceptable test-retest reliability.

This study had some limitations. First, participants in online panels are often more adept with digital technology, potentially excluding older adults who lack access or have poor Internet connections. This could lead to unrepresentative results favoring those who are technologically adept. To mitigate these biases, future studies could combine online sampling with traditional methods such as face-to-face interviews or paper surveys to secure a more representative sample of the Korean population. Second, although the study had an adequate sample size of 270, it was still relatively small to meet the requirements for validity and reliability studies. A minimum of 10-15 participants per item is recommended for factor analysis, suggesting a sample size of 270~405 participants for 27 items (De vet et al., 2011). A larger sample size would improve the accuracy of fit indices and provide more robust insights into the scale’s dimensionality. Future studies should aim for a larger sample size to ensure more accurate results.

Despite these limitations, this is the first study that validates the MCLHB-DRR scale in the general Korean population. A key strength was recruiting a national sample of middle-aged and older adults through an online panel survey, ensuring representativeness. The study also followed the WHO guidelines (2016) for translation and adaptation. Content validity was rigorously assessed

by 13 dementia healthcare experts in occupational therapy, ensuring the scale's validity in the context of dementia healthcare and highlighting its specialized and practical applicability.

The implications of this study are as follows: First, the translated questionnaire is reliable for evaluating middle-aged and older adults' beliefs and attitudes toward lifestyle and health behavioral changes for dementia risk reduction. Further studies should evaluate the relationship between motivation for these changes and actual behavioral change. Assessing beliefs in the Korean community can help develop educational or intervention programs focused on reducing dementia risk. Second, this instrument, based on the HBM, allows varied scores on each subscale, helping researchers tailor interventions to individual motivations. The K-MCLHB-DRR supports designing client-centered intervention plans guided by goals, values, beliefs, and occupational needs (AOTA, 2020), making it valuable in occupational therapy. The evaluation revealed lower perceived susceptibility and cues to action compared with other countries, despite high perceived benefits, barriers, general health motivation, and self-efficacy, indicating the need for educational interventions to enhance risk awareness and motivate behavioral changes, as well as proactive measures such as regular health screenings and continuous communication strategies. In summary, this instrument is crucial for assessing beliefs and attitudes necessary for lifestyle adaptations to reduce dementia risk. It can be effectively utilized in future dementia healthcare studies, including cross-sectional studies to evaluate associations with actual behavior changes, and longitudinal studies should examine changes before and after specific interventions.

Future studies should address the following aspects: due to the broad range of potential

behavioral changes participants could consider, they may not have understood which changes the questions referred to. To address knowledge gaps, future studies should provide preliminary information about dementia-preventing behaviors and lifestyles before participants complete the scale. Furthermore, including culturally relevant examples for each item can enhance comprehension. The validity and reliability of the 26 items have been confirmed. Although item 8 was removed due to cultural inappropriateness, the remaining items were deemed culturally appropriate. Experts suggested providing specific examples for some items. For instance, item 9 states, "it would be more serious for me to develop dementia than if I developed other diseases." Including specifics on how dementia affects health, lifestyle, emotional, cognitive, and quality of life, with examples relevant to Korean culture, would be helpful. A Delphi study may be needed to determine prior knowledge and derive suitable examples. Future studies should refine the 26 items by incorporating examples that reflect Korean cultural and social attributes.

V. Conclusion

The results of this study proved the validity and reliability of the Korean version of the MCLHB-DRR scale as an instrument for assessing attitudes and beliefs regarding lifestyle and health behavioral changes for reducing dementia risk in the Korean population aged ≥ 50 years. This instrument can be effectively used in future studies in occupational therapy, such as cross-sectional studies to investigate its association with actual behavioral changes and longitudinal studies to examine pre-post changes.

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Appendix

Appendix 1. The final Korean Version of the Motivation to Change Lifestyle and Health Behaviors for Dementia Risk Reduction Scale (K-MCLHB-DRR) with Validated Validity and Reliability

한글판 치매위험 감소를 위한 라이프스타일 및 건강행동의 변화동기 척도						
문항		전혀 동의하지 않음	동의하지 않음	보통	동의함	매우 동의함
1	나의 치매 발병 가능성은 크다.					
2	나는 미래에 치매에 걸릴 가능성이 높다고 느낀다.					
3	내가 치매에 걸릴 강력한 가능성이 있다.					
4	나는 향후 10년 이내에 치매에 걸릴 것이다.					
5	나는 치매를 생각하면 겁이 난다.					
6	나는 치매에 관해 생각할 때 심장이 빠르게 뛴다.					
7	내가 치매에 걸리면 내 자신에 대한 느낌이 바뀔 것 같다.					
8	나는 다른 질병에 걸리는 것보다 치매에 걸리는 것이 더 심각할 것 같다.					
9	전문가들로부터 얻는 정보와 조언은 내가 생각해 본 적 없는 것을 제공해 줄 수도 있고, 치매에 걸릴 가능성을 낮출 수도 있다.					
10	라이프스타일과 건강습관을 바꾸는 것은 내가 치매에 걸릴 가능성을 줄이는 데 도움이 된다.					
11	라이프스타일과 건강행동을 바꿈으로써 내가 얻을 수 있는 것이 많다.					
12	더 건강한 라이프스타일과 행동에 적응하는 것은 내가 치매에 걸리는 것을 예방해 줄 것이다.					
13	나는 라이프스타일과 건강습관을 바꾸기에는 너무 바쁘다.					
14	라이프스타일과 행동을 바꾸기에는 나의 재정 상황이 허락되지 않는다.					
15	가족에 대한 책임감은 나의 라이프스타일과 행동을 바꾸기 어렵게 만든다.					
16	라이프스타일과 행동을 바꾸는 것은 나의 일정에 지장을 준다.					
17	나는 건강증이 있어서 라이프스타일과 행동을 바꿔야 한다는 생각이 든다.					
18	나는 치매 위험요인들을 가지고 있어서 라이프스타일과 행동을 바꿔야 한다는 생각이 든다.					
19	나는 대중매체를 통해 치매에 대해 더 알게 되어서 라이프스타일과 행동을 바꿔야 한다는 생각이 든다.					
20	나는 가족 구성원이 치매에 걸린 것을 알게 되어서 라이프스타일과 행동을 바꿔야 한다는 생각이 든다.					
21	나에게 양호한 건강만큼 중요한 것은 없다.					
22	나는 나의 건강에 대해 종종 생각한다.					
23	나는 내 자신의 건강에 관심을 기울여야 한다고 생각한다.					
24	나는 나의 건강을 염려하고 있다.					
25	나는 나의 라이프스타일과 행동을 변화시켜 치매에 걸릴 위험을 줄일 수 있다고 확신한다.					
26	나는 치매에 걸릴 위험을 바꿀 변화를 만들 수 있다.					